

STACK (Method Area)

main(String[] args)

Stack Frame	
Variable	Value
args	(reference to String array)
myCar	0x1001

myCar is a reference variable stored in STACK. It holds the address of the Car object in HEAP.

Legend

0x1001 = Memory Address (Example)

HEAP (Objects Area)

Object: Car (created by new Car())

Address: 0x1001

Field (Attribute)	Value
colour	"Black"
speed	0
engineOn	false
leftIndicatorOn	false
rightIndicatorOn	false

Object States During Execution

1. After new Car()

White
0
false
false
false

2. After setColour("Black")

Black
0
false
false
false

3. After startEngine()

Black
0
true
false
false

4. After accelerate() x2

Black
20
true
false
false

5. After turnLeftIndicatorOn()

Black
20
true
true
false

6. After brake()

Black
10
true
true
false

7. After turnIndicatorsOff()

Black
10
true
false
false

8. After stopEngine()

Black
0
false
false
false

Key Points

- New object (Car) is created in HEAP.
- Reference variable (myCar) is stored in STACK.
- All attributes (fields) of the object live in HEAP.
- Methods operate on the object in HEAP through the reference.